

C++ Project - GymSystem

The GymSystem is a gym management platform that focuses on the interaction between trainers and members, as well as the logistical management of classes and equipment. The system allows for centralized management of gym resources under a single class that manages all entities.

Key Features of the System:

- **User Management:** All users (members and trainers) inherit from the abstract class `Person`, which includes basic personal details.
- **The Trainer's Role:** The trainer is the central entity responsible for the professional aspect. They manage a personal member list (`myMembers`), and only they are authorized to create or update workout plans for them.
- **Class Types:** The system supports two types of sessions: group classes (`GroupSession`) and personal training (`PrivateSession`), each tailored to the participants' needs.
- **Logistical Management:** `GymSystem` is responsible for adding/deleting staff and memberships, creating new classes, and managing equipment inventory and reports.
- **Equipment Hierarchy:** The system demonstrates the use of multiple inheritance. There are devices that are both weights and machines (`weightMachine`), inheriting from both the `weights` and `machines` classes.
- **ClassSession Structure:** For every class (`Session`), there is a direct **Reference** to the `Trainer`. This means that upon creating the class, a specific trainer is assigned to it, and they cannot be changed or replaced by another `Trainer` for that same instance of the class.
- **Workout Plan (WorkoutPlan):** Unlike the other dynamic components, the workout plan inside the `Member` is a regular object (**Composition**) and not a pointer. This ensures that the workout plan is an inseparable part of the member and is created/deleted along with them.
- **Subscription Types and Payments:** The system supports different price calculations for monthly or annual subscriptions and records every payment action into the member's history.

